

Experiment 5

Aim: Analyzing unstructured data using Natural Language Tool Kit.

Theory:

Unstructured data refers to information that does not follow a predefined format or model. Examples include text documents, emails, social media posts, reviews, and news articles. Such data is difficult to analyze directly using traditional data processing techniques.

The **Natural Language Toolkit (NLTK)** is a powerful Python library that provides tools and resources for processing and analyzing unstructured textual data. It enables the conversion of raw text into a structured and meaningful form suitable for computational analysis.

Steps in Analyzing Unstructured Data

- 1. Text Preprocessing:**
Raw text is cleaned by removing unwanted characters, punctuation, and stop words. Tokenization is used to split text into words or sentences.
- 2. Linguistic Analysis:**
NLTK supports Part of Speech tagging, stemming, and lemmatization to understand the grammatical structure and base form of words.
- 3. Semantic Analysis:**
WordNet and other lexical resources are used to analyze word meanings, synonyms, and relationships.
- 4. Feature Extraction:**
Important words and patterns are extracted from text for further analysis and modeling.

Applications of Unstructured Data Analysis

- Sentiment analysis
- Opinion mining
- Text classification
- Spam detection
- Information retrieval

Source Code and Outputs:

```
(1) !pip install -U nltk
✓ 6s
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Collecting nltk
  Downloading nltk-3.9.2-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Downloading nltk-3.9.2-py3-none-any.whl (1.5 MB)
1.5/1.5 MB 20.4 MB/s eta 0:00:00
Installing collected packages: nltk
Attempting uninstall: nltk
  Found existing installation: nltk 3.9.1
  Uninstalling nltk-3.9.1:
    Successfully uninstalled nltk-3.9.1
```

```
Successfully installed nltk-3.9.2
```

```
[2] import nltk
```

```
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```

```
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')
nltk.download('maxent_ne_chunker')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')
```

```
... [nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
```

```
131 import nltk
    from nltk.tokenize import word_tokenize, sent_tokenize
    from nltk.corpus import stopwords
    from nltk.stem import PorterStemmer, WordNetLemmatizer
    from nltk import pos_tag, ne_chunk
    from collections import Counter

    text = """
    Natural Language Processing (NLP) is a field of artificial intelligence that focuses on the interaction between computers and humans through natural language.
    The ultimate goal of NLP is to enable computers to understand, interpret, and generate human language in a way that is valuable.
    For example, NLP is used in applications like chatbots, sentiment analysis, machine translation, and text summarization.
    """

    sentences = sent_tokenize(text)
    words = word_tokenize(text)

    stop_words = set(stopwords.words('english'))
    filtered_words = [word for word in words if word.lower() not in stop_words]

    stemmer = PorterStemmer()
    stemmed_words = [stemmer.stem(word) for word in filtered_words]

    lemmatizer = WordNetLemmatizer()
    lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]

    pos_tags = pos_tag(filtered_words)

    named_entities = ne_chunk(pos_tags)

    word_freq = Counter(filtered_words)

    # Output results
    print("Original Text:", text)
    print("\nTokenized Sentences:", sentences)
    print("\nTokenized Words:", words)
    print("\nFiltered Words (No Stopwords):", filtered_words)
    print("\nStemmed Words:", stemmed_words)
    print("\nLemmatized Words:", lemmatized_words)
    print("\nPOS Tags:", pos_tags)
    print("\nNamed Entities:", named_entities)
    print("\nWord Frequency Analysis:", word_freq.most_common(5))
```

```

... Original Text:
Natural Language Processing (NLP) is a field of artificial intelligence that focuses on the interaction between computers and humans through natural language.
The ultimate goal of NLP is to enable computers to understand, interpret, and generate human language in a way that is valuable.
For example, NLP is used in applications like chatbots, sentiment analysis, machine translation, and text summarization.

Tokenized Sentences: ['\nNatural Language Processing (NLP) is a field of artificial intelligence that focuses on the interaction between computers and humans through natural language.', 'The ul']
Tokenized Words: ['Natural', 'Language', 'Processing', '(', "'", 'NLP', ')', 'is', 'a', 'field', 'of', 'artificial', 'intelligence', 'that', 'focuses', 'on', 'the', 'interaction', 'between', 'comput']
Filtered Words (No Stopwords): ['Natural', 'Language', 'Processing', '(', "'", 'NLP', ')', 'field', 'artificial', 'intelligence', 'focuses', 'interaction', 'computers', 'humans', 'natural', 'language']
Stemmed Words: ['natur', 'languag', 'process', '(', "'", 'nlp', ')', 'field', 'artifici', 'intellig', 'focus', 'interact', 'comput', 'human', 'natur', 'languag', '.', 'ultim', 'goal', 'nlp', 'enabl']
Lemmatized Words: ['Natural', 'Language', 'Processing', '(', "'", 'NLP', ')', 'field', 'artificial', 'intelligence', 'focus', 'interaction', 'computer', 'human', 'natural', 'language', '.', 'ultimate']
POS Tags: [('Natural', 'JJ'), ('Language', 'NNP'), ('Processing', 'NNP'), ('(', '(', 'NLP', 'NNP'), (',', ','), ('field', 'NN'), ('artificial', 'JJ'), ('intelligence', 'NN'), ('focuses', 'VBZ')
Named Entities: {}
Natural/JJ
Language/NNP
Processing/NNP
/(
(ORGANIZATION NLP/NNP)
)/
field/NN
artificial/JJ
intelligence/NN
focuses/VBZ
interaction/NN
computers/NNs
humans/VBZ
natural/JJ
language/NN
./
ultimate/JJ
goal/NN
(ORGANIZATION NLP/NNP)
enable/JJ

```

```
language/NN
./,
ultimate/JJ
goal/NN
(ORGANIZATION NLP/NNP)
enable/JJ
computers/NN$
understand/VBP
./,
interpret/JJ
./,
generate/JJ
human/JJ
language/NN
way/NN
valuable/JJ
./,
example/NN
./,
(ORGANIZATION NLP/NNP)
used/VBD
applications/NN$
like/IN
chatbots/NN$
./,
sentiment/NN
analysis/NN
./,
machine/NN
translation/NN
./,
text/JJ
summarization/NN
./,

Word Frequency Analysis: [(',', 6), ('NLP', 3), ('.', 3), ('computers', 2), ('language', 2)]
```

Learning Outcomes:

I am able to analyze unstructured textual data using NLTK by applying preprocessing techniques such as tokenization, normalization, and tagging, and extracting meaningful patterns, relationships, and insights from raw text for real-world applications.